

AI Based Multi Cancer Diagnosis System

Project Team

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Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Module 1:Cancer Detection and Segmentation	2
1.3.2	Module 2: Cancer Severity and Risk Prediction	3
1.3.3	Module 3: Web Application Interface	3
1.3.4	Module 4: Reporting and Collaboration	3
1.3.5	Module 5: Integration and Security	4
1.4	User Classes and Characteristics	4
2	Project Requirements	7
2.1	Use-Cases	7
2.2	Functional Requirements	8
2.2.1	Module 1: Cancer Detection and Segmentation	9
2.2.2	Module 2: Cancer Severity and Risk Prediction	9
2.2.3	Module 3: Web Application Interface	9
2.2.4	Module 4: Reporting and Collaboration	10
2.2.5	Module 5: Integration and Security	10
2.3	Non-Functional Requirements	10
2.3.1	Usability	10
2.3.2	Performance	10
2.3.3	Security	11
2.4	Use-Case Diagram	11
2.5	Domain Model	12
3	System Overview	13
3.1	Architectural Design	13
3.1.1	Architecture Diagram	13
3.2	Design Models	14
3.2.1	Activity Diagram 01	14
3.2.2	Activity Diagram 02	15
3.2.3	Activity Diagram 03	16

3.2.4	Activity Diagram 04	17
3.3	Data Flow Diagrams	18
3.3.1	DFD Level 0	18
3.3.2	DFD Level 1	19
3.3.3	DFD Level 2 D1	20
3.3.4	DFD Level 2 D2	21
3.3.5	DFD Level 2 D3	22
3.3.6	DFD Level 2 D4	23
3.3.7	DFD Level 2 D5	24
3.4	System Sequence Diagram	25
3.5	Data Design	26
3.5.1	Medical Professional Schema	26
3.5.2	Security Schema	26
3.5.3	Medical Image Schema	26
3.5.4	Cancer Detection Schema	26
3.5.5	Cancer Segmentation Result Schema	27
3.5.6	Risk Prediction Schema	27
3.5.7	Diagnostic Report Schema	28
3.5.8	Collaboration Schema	28
References		29

List of Figures

2.1	Use Case Diagram	11
2.2	Domain Model	12
3.1	Architecture Diagram	13
3.2	Activity Diagram	14
3.3	Activity Diagram	15
3.4	Activity Diagram	16
3.5	Activity Diagram	17
3.6	DFD-Level0	18
3.7	DFD-Level1	19
3.8	DFD-Level2	20
3.9	DFD-Level2	21
3.10	DFD-Level2	22
3.11	DFD-Level2	23
3.12	DFD-Level2	24
3.13	System Sequence Diagram	25

List of Tables

Chapter 1

Introduction

This project proposal document outlines the development of an AI-based multi-cancer detection system that specializes in lung, breast, and skin cancers. The system will employ advanced deep learning models to perform cancer classification, segmentation, nodule size prediction, cancer severity assessment, and risk prediction. The project aims to create a comprehensive platform for medical professionals that not only enhances diagnostic accuracy but also provides an interactive web application for seamless user experience. The document provides a detailed background of the system, identifies the problem being addressed, discusses the motivation behind the project, and outlines the solution approach and key stakeholders.

1.1 Problem Statement

The proposed project mainly focuses on the development of an AI-based system for the detection and segmentation of cancer in the lung, breast, and skin. More precisely, the higher-order deep learning techniques in CNN and Transformer will be utilized in the proposed scheme to accurately segment the tumor region from medical images. It considers CT scan images in the case of lung cancer, mammogram images in the case of breast cancer, and 2D images in the case of skin cancer. The system will also predict the size of the cancerous nodules or lesions using radiomic features and, further based on those features, predict the risk of advancement of cancer in the lungs. The interface will be web-based and will interact with the models, having visualized detection of cancerous areas, collaborating on it with other colleagues, and finally comprehensive diagnostic reports by the medical personnel. The platform shall include 3D visualization for interaction with CT scan images, security features for patient record keeping, data encryption, and APIs for integrating the system with hospital management systems and EHRs. This will lead to better diagnostics, less friction in workflows, and closer collaboration among

professionals for improved patient outcomes.

1.2 Scope

The scope of this project is to develop an AI-based system for the detection and segmentation of lung, breast, and skin cancers. The system will employ advanced deep learning techniques such as Convolutional Neural Networks (CNNs) and Transformers to perform accurate segmentation of cancerous regions in medical images, specifically using CT scans for lung cancer, mammograms for breast cancer, and 2D images for skin cancer. Additionally, the system will estimate the size of cancerous nodules or lesions using radiomic features and predict the risk of cancer progression in the lungs. The web-based interface will enable medical professionals to interact with the models, view detected cancerous areas, collaborate with colleagues, and generate comprehensive diagnostic reports. The platform will also support 3D visualization tools for interacting with CT scan images, secure patient record-keeping, data encryption, and APIs for integration with hospital management systems and electronic health records (EHR). The solution aims to enhance diagnostic accuracy, streamline workflows, and foster collaboration among healthcare professionals, thereby improving patient outcomes.

1.3 Modules

The system is divided into various modules for different functionalities and features in the multi-cancer detection system. The modules are organized regarding the types of cancer detection and extra functionality, which is provided by the system.

1.3.1 Module 1: Cancer Detection and Segmentation

In this module, models will be applied to detect and segment lung, breast, and skin cancers, each based on various kinds of imaging techniques with different machine learning models.

Lung Cancer Detection and Segmentation: Utilize CT Scans for the detection and segmentation of lungs, further segmenting the cancerous regions in order to train the models on nodule size estimation using the detected radiomic features.

Breast Cancer Detection and Segmentation: Breast cancer detection and segmentation by mammograms, estimation of size of nodules with the use of radiomic features.

Skin Cancer Classification and Segmentation: Melanoma, basal cell carcinoma, varieties of skin cancer are to be classified and segmented from 2D skin images .

1.3.2 Module 2: Cancer Severity and Risk Prediction

It provides the prediction about the severity in lung cancer and estimates the risk for further development of cancer .

Lung Cancer Severity Prediction:It will predict the severity of the detected lung cancer, which would further facilitate finding the urgency and treatment planning.

Cancer Growth Risk Prediction:In the case of lung cancer, it calculates the possibility of growth and dispersion of cancer because in the near future, it would be possible to manage such a risk .

1.3.3 Module 3: Web Application Interface

This module represents the development of the Web application through which the user can interact with it by generating reports and is able to collaborate in it .

User Dashboard:Implement an active, user-friendly interface through which medical professionals will be able to upload images, visualize model outputs, and manage cases via the frontend .

3D Visualization Tools:Give the user the ability to interactively visualize the CT scan images in 3D in a manner that could further help illustrate the understanding of the segmented regions .

1.3.4 Module 4: Reporting and Collaboration

This module will also include report generation and the facility for health professionals to collaborate on a specific case .

1. **Report Generation Tools:** Detailed analysis reports in PDF format, comprising the predictions with segmentation regarding size/location of the nodules .
2. **Collaboration Features:** The user can be allowed to make invitations for other medical experts so as to collaborate on particular cases in terms of discussing the findings.

1.3.5 Module 5: Integration and Security

This module will work on data security and its integration with other systems .

1. **Data Encryption:** Avail different methods of data encryption to integrate the data and keep it safe during transport and at rest.
2. **API Development for Integration:** Development of APIs for seamless integration with Hospital Management Systems, EHRs, Third-party software.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Radiologists	Specialists who will focus on the analysis of image data, ensuring the accuracy of tumor segmentation and classification, particularly in lung and breast cancer cases.
Oncologists	Cancer specialists who will use the system for evaluating the severity and progression of cancer and making treatment recommendations based on risk predictions.
Medical Researchers	Researchers using the system to collect and analyze data to improve cancer detection algorithms, often collaborating with healthcare professionals to validate findings.
Hospital Management	The administration staff who will use APIs to integrate the system with hospital management systems (HMS) and ensure that the system is operating within compliance.
Healthcare Providers	Entities like clinics or hospitals who will be responsible for managing patient records and ensuring that the system aligns with their health information workflows.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-Cases

- **Actors:** Radiologists, Oncologists,
- **Primary Use Cases:**
 - Upload medical images (CT scans, mammograms, 2D skin images)
 - Detect Cancer
 - Visualize cancer segmentation results
 - Generate Diagnostic reports
 - Collaborate with other medical professionals
- **Use Case 1: Upload Medical Images**
 - Actor: Radiologist
 - Description: The user uploads CT scans, mammograms, or 2D skin images for analysis.
 - Precondition: The user is authenticated and logged into the system.
 - Postcondition: The uploaded images are processed and analyzed by the system.

- **Use Case 2: Visualize Cancer Segmentation**

- Actor: Radiologist
- Description: The user visualizes the segmentation of cancerous regions in the medical images uploaded.
- Precondition: The user has uploaded the required images.
- Postcondition: The system displays the segmented areas, with options for 3D visualization for CT scans.

- **Use Case 3: Generate Reports**

- Actor: Radiologist ,Oncologist
- Description: The user generates detailed diagnostic reports based on the segmentation and prediction results.
- Precondition: The user has completed the cancer segmentation process.
- Postcondition: The system generates a PDF report for the patient's case.

- **Use Case 4: Collaborate on Cases**

- Actor:Radiologist, Oncologist
- Description: The user invites other medical professionals to collaborate on the case and discuss the findings.
- Precondition: A report or case has been generated.
- Postcondition: Multiple users are granted access to the case for collaboration.

2.2 Functional Requirements

This chapter presents the functional requirements for the AI-Based Cancer Diagnosis System.

2.2.1 Module 1: Cancer Detection and Segmentation

Following are the requirements for module 1:

1. **FR1:** The system shall detect and segment lung cancer from CT scan images.
2. **FR2:** The system shall detect and segment breast cancer from mammogram images.
3. **FR3:** The system shall classify and segment skin cancers, including melanoma and basal cell carcinoma, using 2D skin images.
4. **FR4:** The system shall allow users to upload medical images (CT, mammogram, or 2D skin images) for processing.
5. **FR5:** The system shall estimate the size of detected nodules or lesions..

2.2.2 Module 2: Cancer Severity and Risk Prediction

Following are the requirements for module 2:

1. **FR1:** The system shall predict the severity of lung cancer based on the size and spread of nodules detect.
2. **FR2:** The system shall assess the risk of lung cancer growth and spread from CT scan images.

2.2.3 Module 3: Web Application Interface

Following are the requirements for module 3:

1. **FR1:** The web application shall provide a dashboard for users to upload images and visualize the segmentation results.
2. **FR2:** The system shall support interactive 3D visualization of CT scan images for better understanding of segmented cancer regions.
3. **FR3:** The system shall allow users to download and share detailed reports in PDF format.
4. **FR4:** The system shall allow users to collaborate by inviting other professionals to review cases.

2.2.4 Module 4: Reporting and Collaboration

Following are the requirements for module 4:

1. **FR1:** The system shall generate diagnostic reports in PDF format summarizing the segmentation and prediction results. [1]
2. **FR2:** The system shall allow users to invite other professionals to collaborate on a specific case.

2.2.5 Module 5: Integration and Security

Following are the requirements for module 5:

1. **FR1:** The system shall use data Hashing methods to ensure secure storage and transmission of patient data.
2. **FR2:** The system shall provide REST APIs for integration with hospital management systems and EHRs.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements.

2.3.1 Usability

- **NFR1:** The system shall allow a user to upload medical images in less than 2 interactions.
- **NFR2:** The dashboard shall be designed to minimize errors, with recovery options for users to retry failed image uploads.

2.3.2 Performance

- **NFR1:** The system shall perform detection and segmentation tasks within 1 minute.
- **NFR2:** The 3D visualization interface shall render CT scans in less than 10 seconds.

2.3.3 Security

- **NFR1:** The system shall require multi-factor authentication for all medical professionals.
- **NFR2:** The system shall encrypt all stored data using hashing algorithms, ensuring the highest levels of protection against unauthorized access.

2.4 Use-Case Diagram



Figure 2.1: Use Case Diagram

2.5 Domain Model

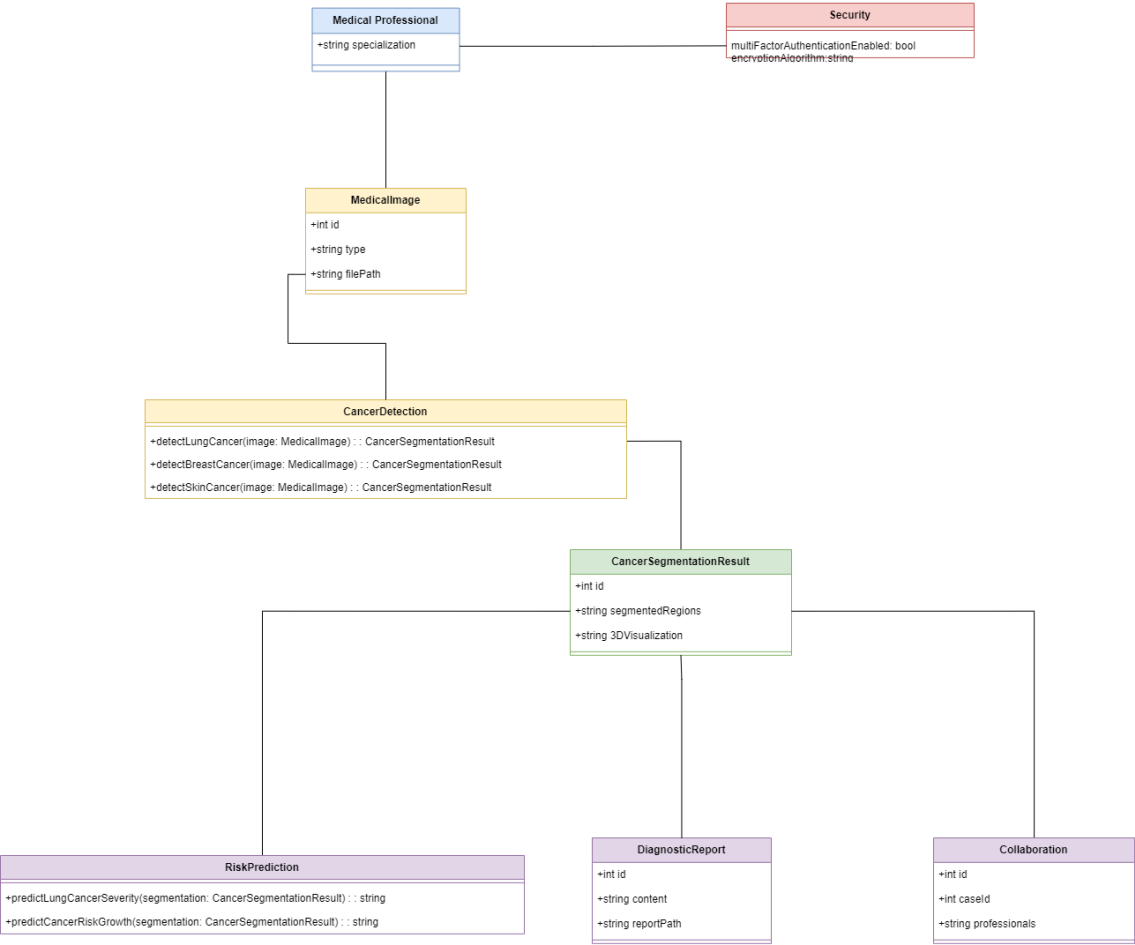


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

3.1.1 Architecture Diagram

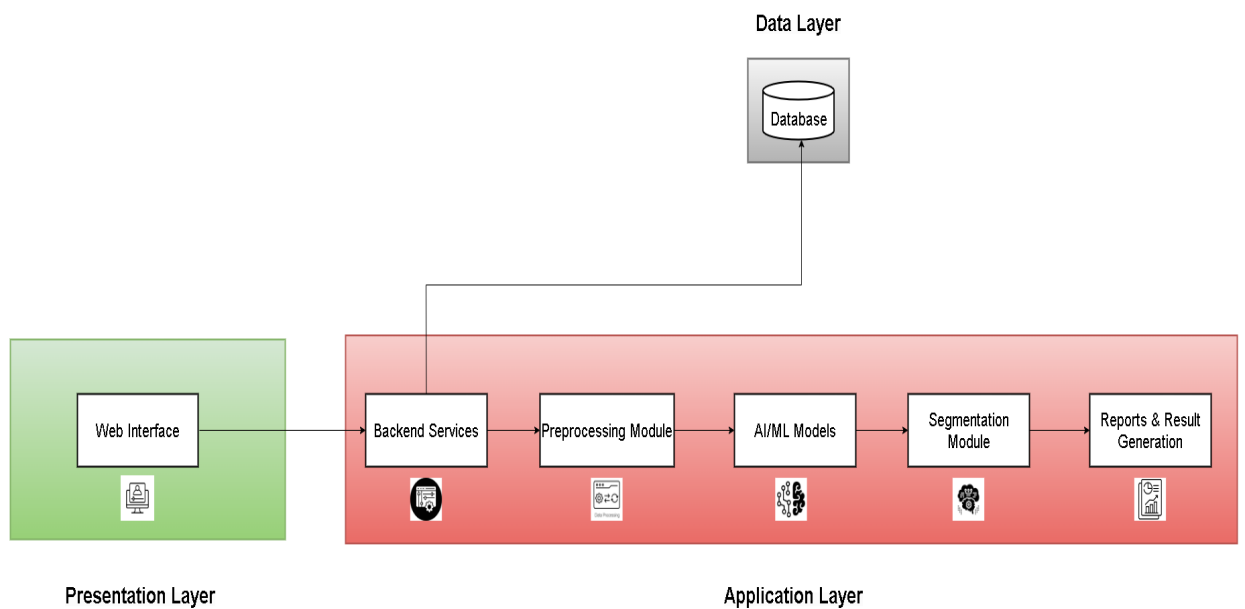


Figure 3.1: Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagram 01



Figure 3.2: Activity Diagram

3.2.2 Activity Diagram 02

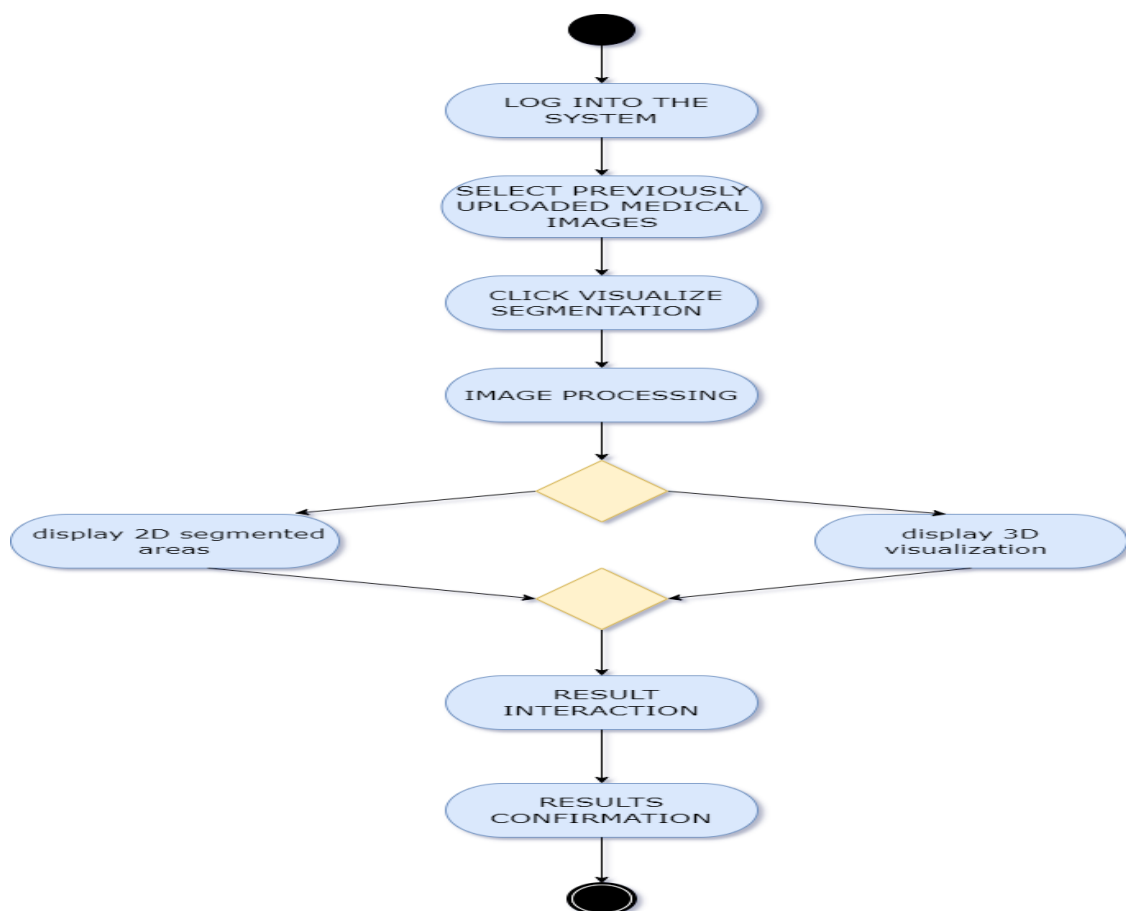


Figure 3.3: Activity Diagram

3.2.3 Activity Diagram 03

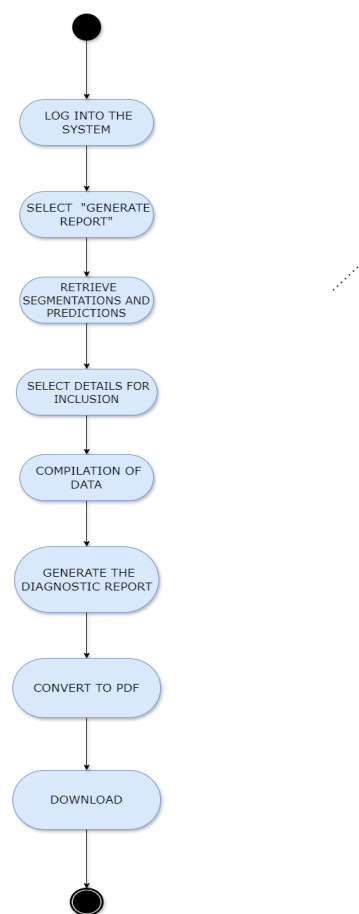


Figure 3.4: Activity Diagram

3.2.4 Activity Diagram 04

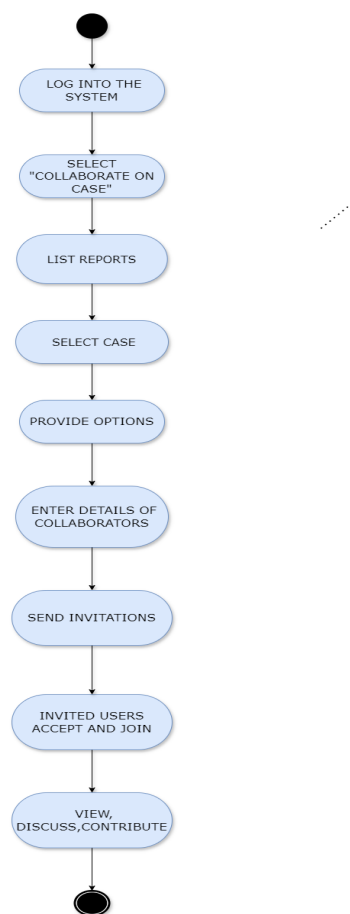


Figure 3.5: Activity Diagram

3.3 Data Flow Diagrams

3.3.1 DFD Level 0

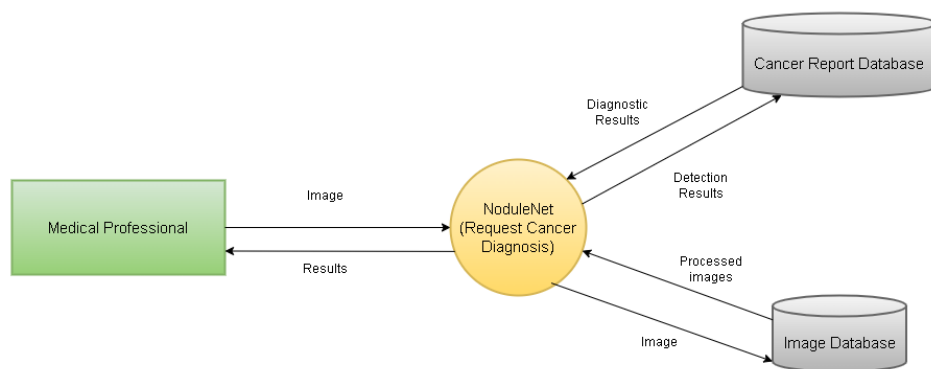


Figure 3.6: DFD-Level0

3.3.2 DFD Level 1

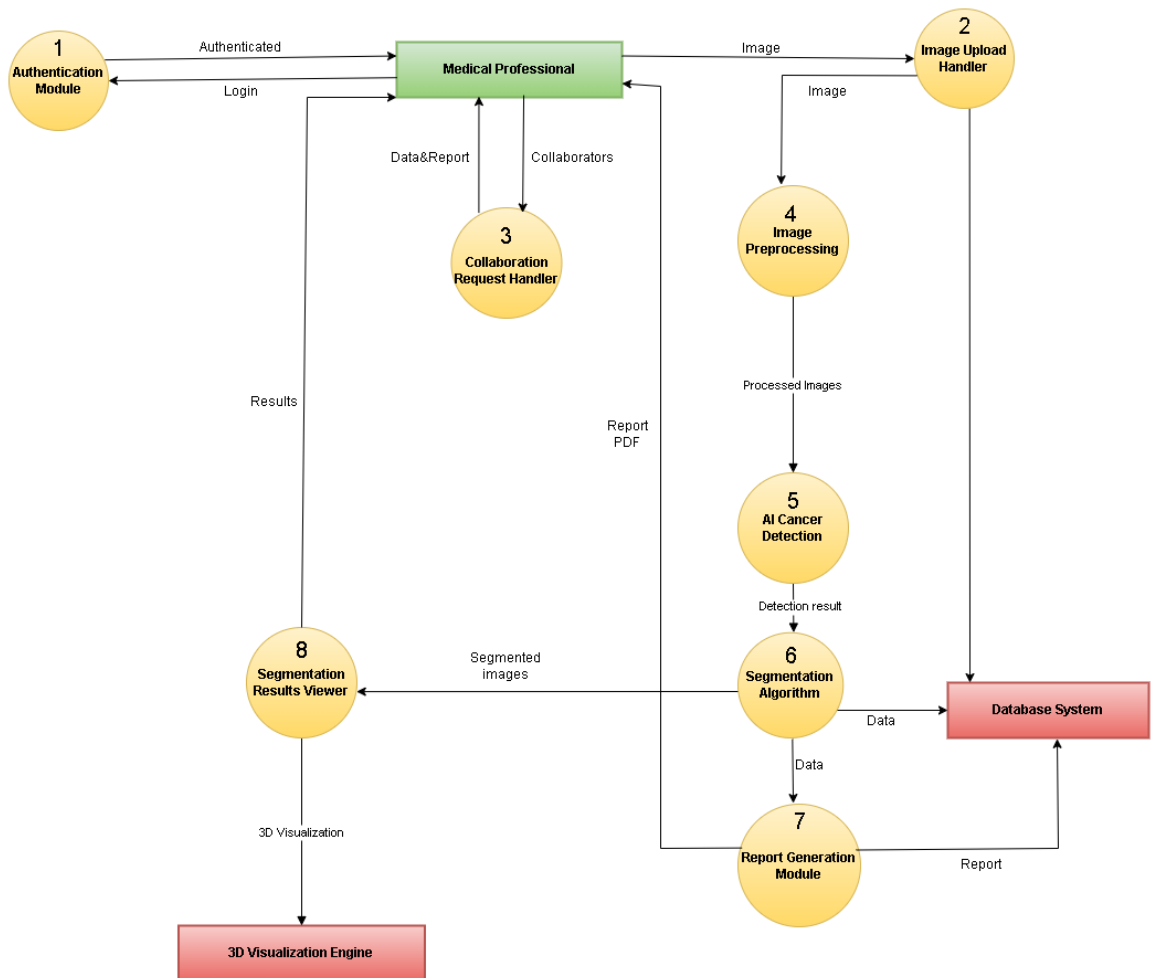


Figure 3.7: DFD-Level1

3.3.3 DFD Level 2 D1

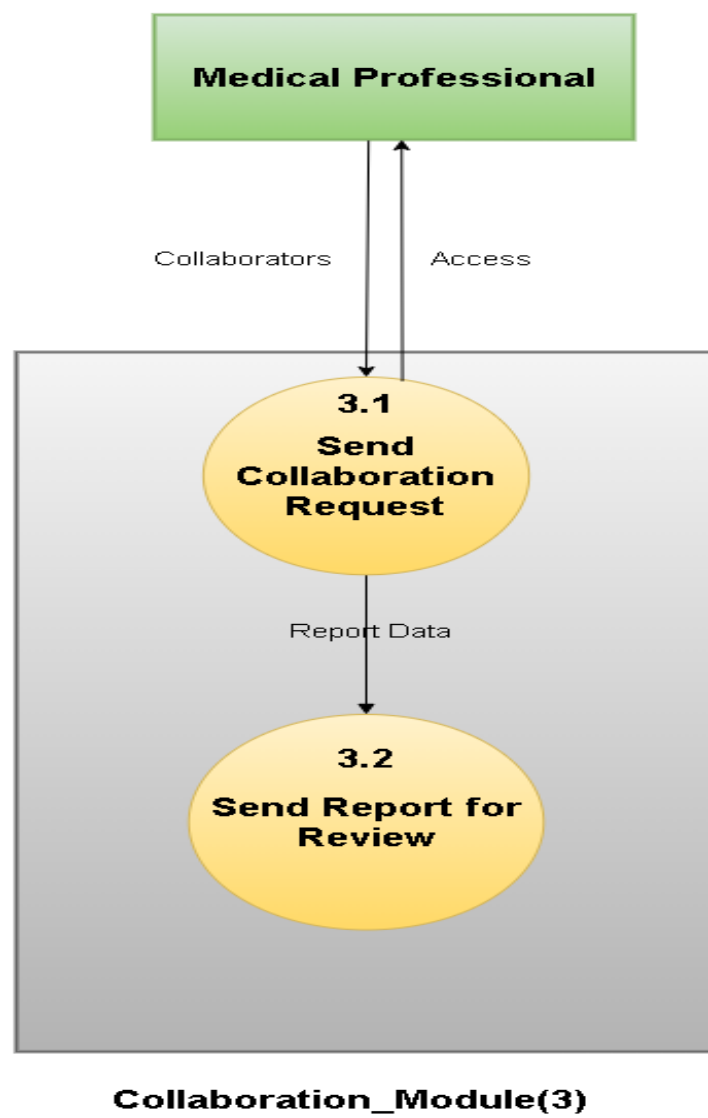


Figure 3.8: DFD-Level2

3.3.4 DFD Level 2 D2

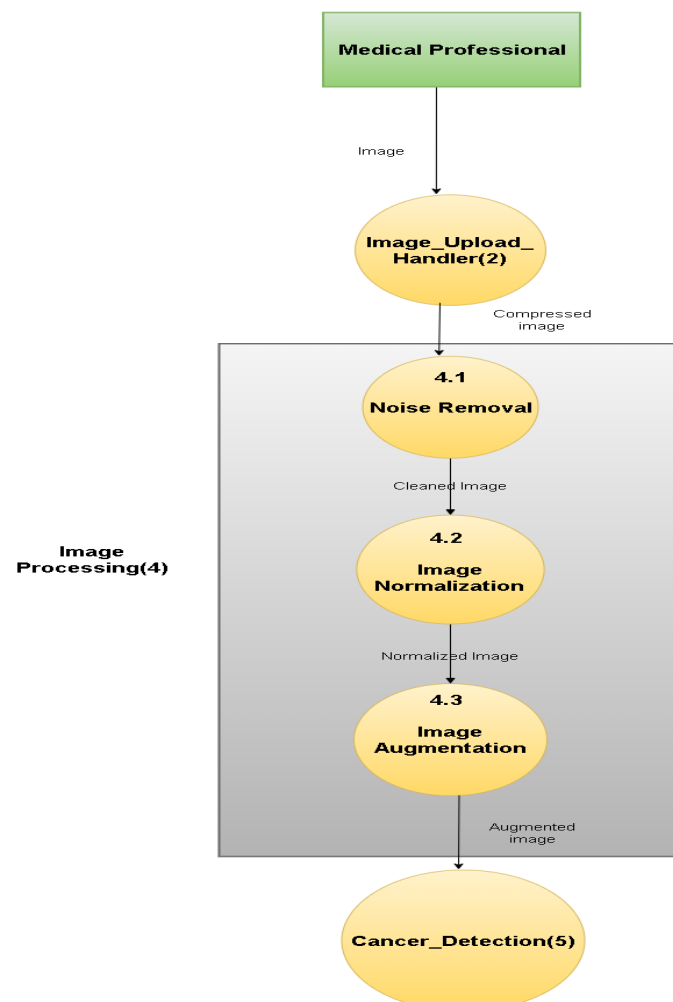


Figure 3.9: DFD-Level2

3.3.5 DFD Level 2 D3

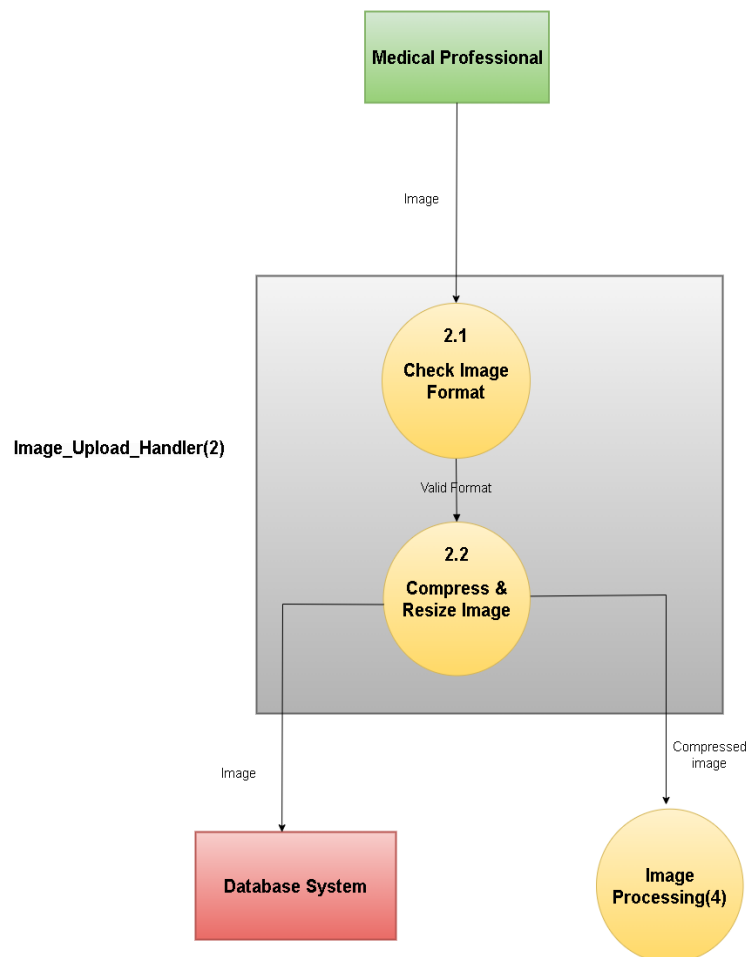


Figure 3.10: DFD-Level2

3.3.6 DFD Level 2 D4

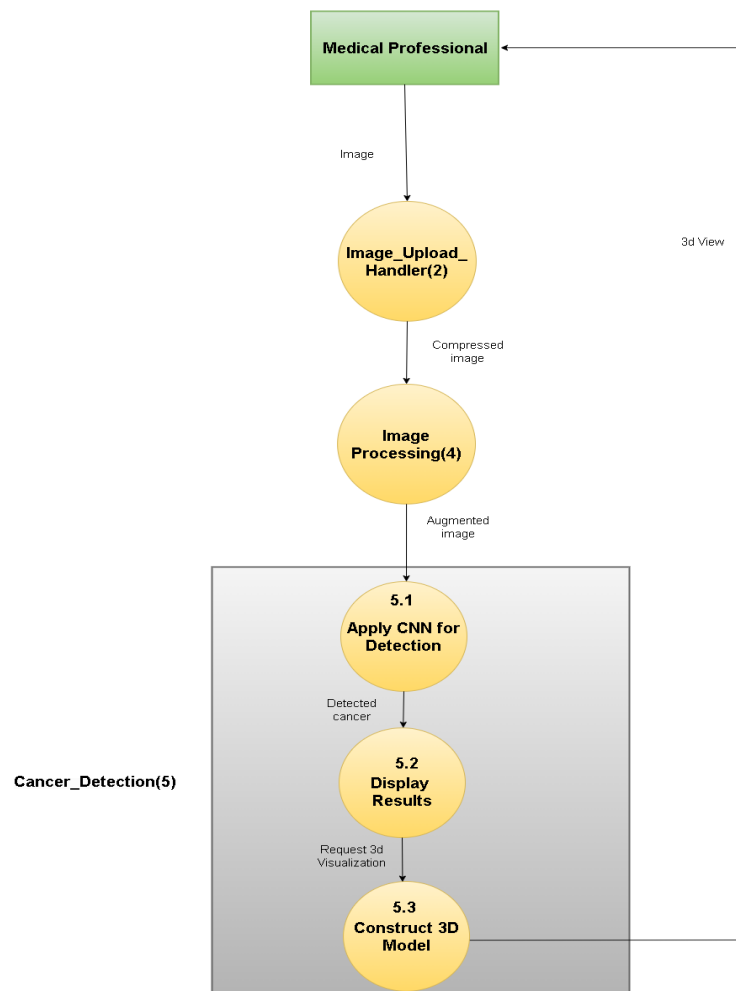


Figure 3.11: DFD-Level2

3.3.7 DFD Level 2 D5

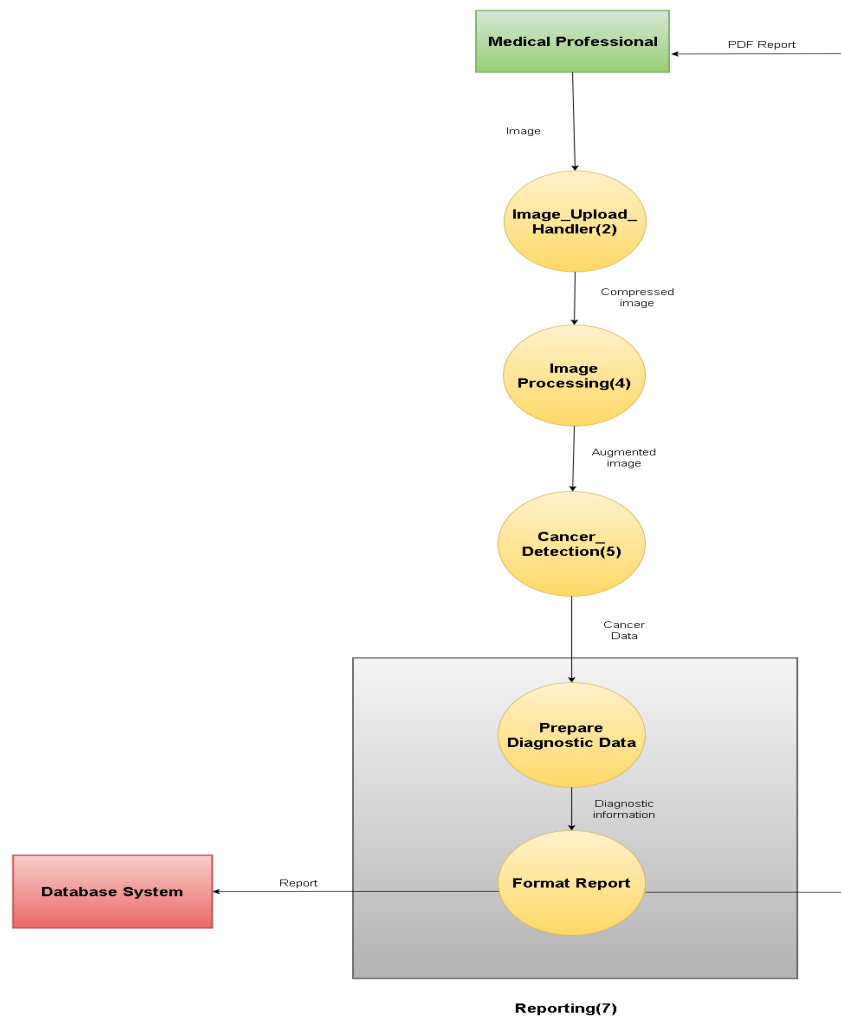


Figure 3.12: DFD-Level2

3.4 System Sequence Diagram

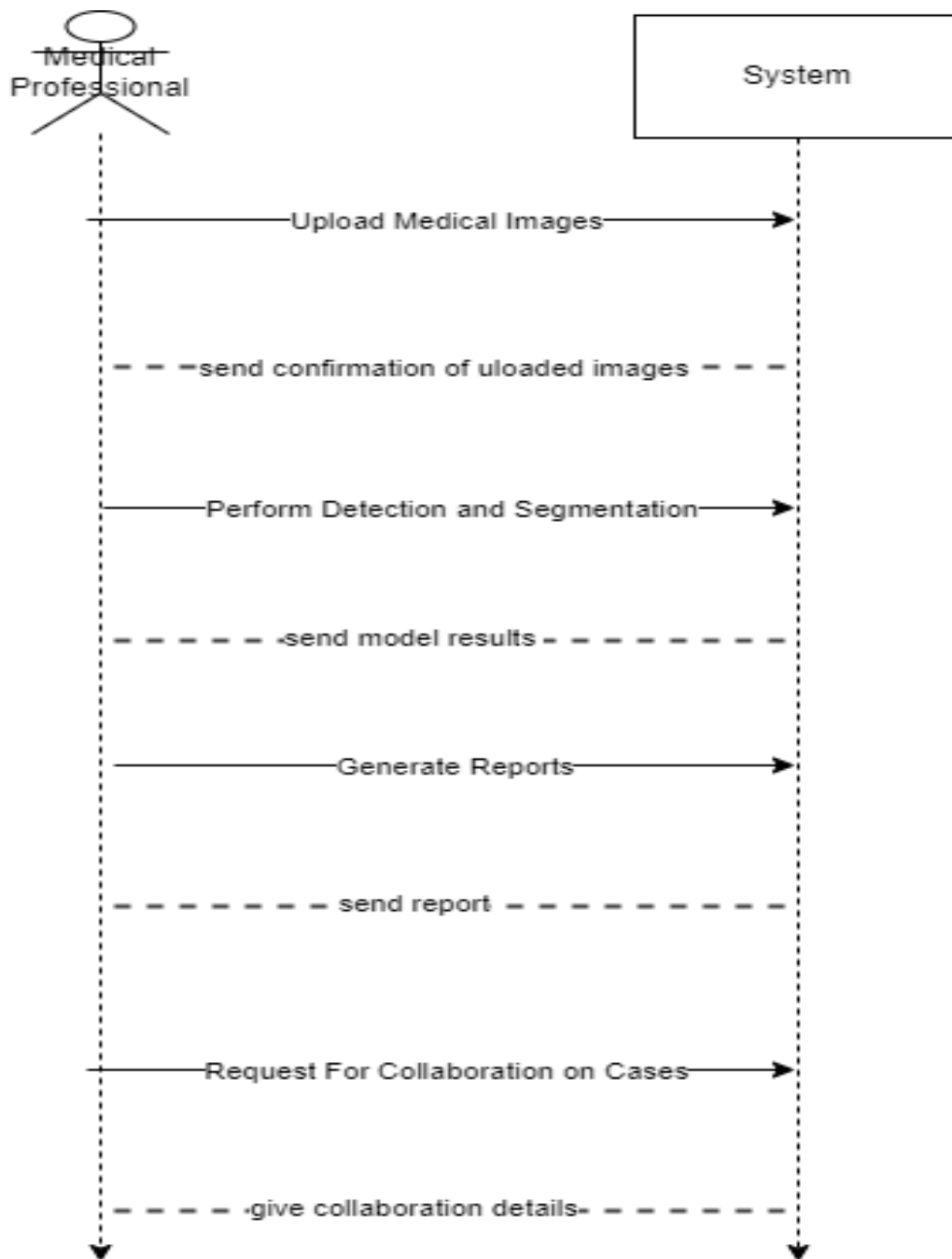


Figure 3.13: System Sequence Diagram

3.5 Data Design

This section outlines the data organization and schemas utilized within the system. The system manages various types of data essential for cancer detection and diagnosis, including medical images, patient information, cancer diagnosis results, segmentation details, and records of medical professionals. Each type of data is structured in specific JSON documents to facilitate efficient access, processing, and management.

3.5.1 Medical Professional Schema

```
{
  "medicalProfessionalId": "doctor_001",
  "specialization": "Oncologist"
}
```

3.5.2 Security Schema

```
{
  "securityId": "sec_001",
  "multiFactorAuthentication": true,
  "encryptedData": "encrypted_string_value"
}
```

3.5.3 Medical Image Schema

```
{
  "imageId": 1,
  "type": "Mammogram",
  "filePath": "https://example.com/images/scan_001.dcm",
  "uploadImage": "function to upload an image"
}
```

3.5.4 Cancer Detection Schema

```
{
  "detectionId": 1,
```

```

"lungCancerDetection": {
  "imageId": 1,
  "segmentationResult": {
    "segmentedRegions": "regions data",
    "3DVisualization": "3d_visualization_data"
  }
},
"breastCancerDetection": {
  "imageId": 2,
  "segmentationResult": {
    "segmentedRegions": "regions data",
    "3DVisualization": "3d_visualization_data"
  }
},
"skinCancerDetection": {
  "imageId": 3,
  "segmentationResult": {
    "segmentedRegions": "regions data",
    "3DVisualization": "3d_visualization_data"
  }
}
}

```

3.5.5 Cancer Segmentation Result Schema

```

{
  "segmentationResultId": 1,
  "segmentedRegions": "regions data",
  "3DVisualization": "3d_visualization_data",
  "visualizeSegmentation": "function to visualize segmentation"
}

```

3.5.6 Risk Prediction Schema

```

{
  "riskPredictionId": 1,
  "lungCancerRiskSeverity": {

```

```
    "segmentationResultId": 1,
    "riskLevel": "High"
  },
  "cancerRiskGrowth": {
    "segmentationResultId": 2,
    "growthRate": "Fast"
  }
}
```

3.5.7 Diagnostic Report Schema

```
{
  "reportId": 1,
  "content": "This is a diagnostic report",
  "reportPath": "https://example.com/reports/report_001.pdf",
  "generateReport": "function to generate report"
}
```

3.5.8 Collaboration Schema

```
{
  "collaborationId": 1,
  "caseId": 1,
  "professionals": [
    "doctor_001",
    "doctor_002"
  ],
  "inviteCollaborators": "function to invite collaborators"
}
```

Bibliography